

ARC-SPRUCE: Post-Quantum Blind Signatures and Anonymous Rate-Limited Credentials from Lattices

ABSTRACT

Blind signatures and anonymous credentials have recently seen a large interest from practitioners due to their usefulness in building privacy-preserving online authentication systems such as the EU Digital Identity Wallet. However, all currently known and practically efficient systems are not secure against quantum computers.

In this work, we use recent lattice-based techniques to modularly design post-quantum blind signatures and anonymous rate-limiting credentials. By using a state-of-the-art Non-Interactive Zero-Knowledge Proof system called SmallWood combined with a PRF based on the Poseidon Hash function, we obtain signature sizes that outperform existing constructions. In addition, our implementation shows competitive computational performance on modest hardware. We also provide the first, to our knowledge, implementation of the SmallWood proof system, which is of independent interest.

CCS CONCEPTS

• Security and privacy → Digital signatures; Privacy-preserving protocols; Pseudonymity, anonymity and untraceability.

KEYWORDS

Blind Signatures, Anonymous Credentials, Post-Quantum Cryptography

ACM Reference Format:

. 2026. ARC-SPRUCE: Post-Quantum Blind Signatures and Anonymous Rate-Limited Credentials from Lattices. In *Proceedings of ACM Conference (Conference'17)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 INTRODUCTION

Anonymous credentials (AC) [Cha85, Bra95, CL01] allow parties to convey certified information about themselves in a privacy-preserving way. The usual roles are issuers, holders, and verifiers. A holder obtains attributes certified by an issuer and later proves statements about those attributes to a verifier without revealing the credential itself. The central security goals are unforgeability and unlinkability: verifiers should only accept credentials derived from an issuer, while issuers and verifiers should not be able to link a presentation to an issuance session or to another presentation.

Certain AC use cases, such as anonymous rate limiting [YWFH26, SK25], require that a credential be usable only a bounded number

of times in a given context, for example per epoch or per service. Existing classical constructions can provide compact credentials and presentations, but practically efficient post-quantum ACs remain much less developed. This gap is relevant for digital identity systems, including wallet deployments that will eventually require post-quantum migration.

Building ACs. Anonymous credentials are often built from blind signatures. The holder first commits to the message to be signed, the issuer signs a target derived from this committed message, and the holder later presents a zero-knowledge proof of knowledge of a valid message-signature pair. A generic post-quantum combination of standard signatures, commitments, and zero-knowledge proofs is currently too large for the settings targeted here. We therefore combine a proof-friendly lattice signature, an Ajtai-style commitment, and SmallWood zero-knowledge proofs in a relation that is designed to be compact.

1.1 Contributions

We construct a post-quantum blind-signature and anonymous rate-limited credential design from lattice assumptions and hash-based zero knowledge. The construction builds on proof-friendly lattice signatures from vanishing SIS with rational hashing [DKLW25a, DKLW25b], NTRU-style trapdoor sampling for hash-and-sign signatures [ENS⁺23], and the SmallWood proof system for relatively small statements [FR25]. Rate limiting is implemented by signing a hidden PRF key and proving correct tag derivation during presentation.

The blind-issuance layer uses an Ajtai commitment and a two-party randomness step. The holder commits to the full signed message and its holder-side randomness before the issuer sends its contribution. The final rational-hash randomness is obtained by centering the sum of the two contributions. This preserves the old shared-randomness issuance architecture, while the rational hash itself is the BB-tran form of DKLW:

$$T = B_0 + B_1\mu + B_2r_0 + Z, \quad (B_3 - r_1) \odot Z = 1.$$

The target is computed from the bare hidden witness elements, not from the commitment value. The commitment serves to bind the holder to its message and holder-side randomness before the issuer randomness is known.

1.2 Technical overview

The full signed message is

$$\mu = \mathbf{m} \parallel \mathbf{k},$$

where $\mathbf{m}$ is the attribute vector and $\mathbf{k}$ is the secret PRF key.

Issuance. The holder samples $\mathbf{k}$, holder-side hash randomness r_0^H, r_1^H , and commitment-only randomness $\bar{r}$. It sends the Ajtai commitment

$$\mathbf{c} = \mathbf{A}_{\text{Com}} [\mu \parallel r_0^H \parallel r_1^H \parallel \bar{r}]^\top$$

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Conference'17, July 2017, Washington, DC, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM...\$15.00

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

to the issuer. The issuer samples bounded randomness r_0^I, r_1^I and returns it. The holder computes

$$r_0 = \text{center}(r_0^H + r_0^I), \quad r_1 = \text{center}(r_1^H + r_1^I),$$

then derives

$$Z = (B_3 - r_1)^{-1}, \quad T = B_0 + B_1\mu + B_2r_0 + Z.$$

The holder sends T together with a pre-sign NIZK proving that the same bounded witness opens the commitment, computes the centered randomness, and satisfies the BB-tran target relation. If the proof verifies, the issuer samples a short preimage $\mathbf{u} \leftarrow \text{SampPre}(\text{td}, T)$ such that $\mathbf{A}\mathbf{u} = T$ and returns it to the holder.

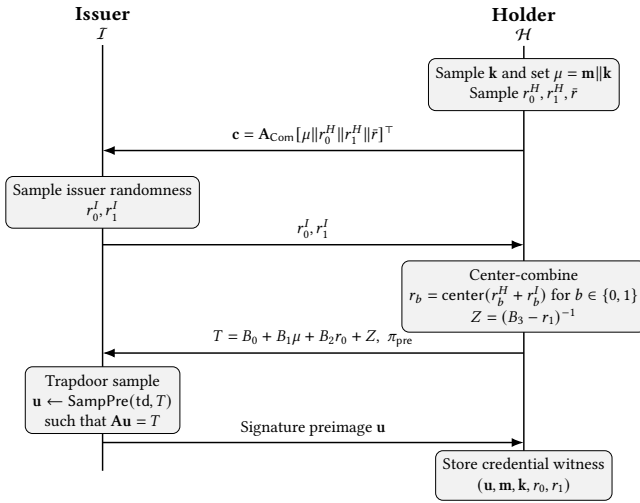


Figure 1: Issuance flow. The pre-sign proof binds the BB-tran target to the committed message and the centered shared randomness, while hiding the holder witness.

Showing. To present a credential, the holder chooses a nonce, computes $\text{tag} = F(\mathbf{k}, \text{nonce})$, and proves knowledge of $\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z$ such that

$$\begin{aligned} (B_3 - r_1) \odot Z &= 1, \\ \mathbf{A}\mathbf{u} &= B_0 + B_1(\mathbf{m}||\mathbf{k}) + B_2r_0 + Z, \\ \text{tag} &= F(\mathbf{k}, \text{nonce}). \end{aligned}$$

The verifier sees only $(\text{nonce}, \text{tag}, \pi_{\text{show}})$ and enforces the rate policy by recording accepted $(\text{nonce}, \text{tag})$ pairs.

SmallWood proofs. We use SmallWood [FR25] as the NIZK for the issuance and showing relations. Both relations are expressed over committed witness rows packed on an evaluation domain. The issuance proof checks the Ajtai opening, the centering equations, and the BB-tran target relation. The showing proof adds the short preimage equation and the PRF-tag relation. The exact compiled statements are stated in Section 5; proof-system details are deferred to Appendix C.

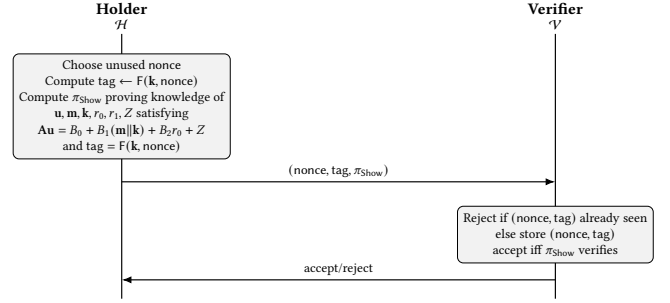


Figure 2: Showing flow. The tag is deterministic for a fixed key and nonce, enabling stateful rate limiting, while the proof hides the credential witness.

2 PRELIMINARIES

Notation. Let λ denote the security parameter and let $\text{negl}(\lambda)$ be any positive function negligible in λ . We write PPT for probabilistic polynomial time. For integers $a < b$, write $[a, b] = \{a, a+1, \dots, b\}$ and use $[b]$ as shorthand for $[1, b]$. Vectors are written in bold lower-case letters and matrices in bold upper-case letters.

We work over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ for power-of-two N and its quotient $R_q = R/qR$, identified with polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$. For $a \in \mathbb{Z}$, let $\langle a \rangle_q \in \{0, \dots, q-1\}$ denote reduction modulo q . When a field element represents a signed value, we use the centered representative. For $x \in R_q$, the notation $\|x\|_\infty \leq B$ means that every coefficient of x lies in $[-B, B]$; the same convention is applied component-wise to vectors over R_q .

The full signed message is written

$$\mu = \mathbf{m}||\mathbf{k},$$

where $\mathbf{m}$ encodes holder attributes and $\mathbf{k}$ is the secret PRF key used for rate limiting. The public coefficient bound for message and hash randomness is denoted β_{msg} .

2.1 Commitment schemes and Ajtai commitments

Definition 2.1 (Commitment scheme). A non-interactive commitment scheme is a tuple of PPT algorithms

$$\text{Com} = (\text{Setup}, \text{Commit}, \text{Verify})$$

with the following syntax. The setup algorithm outputs public parameters pp_{Com} ; the commit algorithm on input a message m outputs a commitment c and an opening d ; and $\text{Verify}(c, m, d)$ deterministically checks validity of the opening. Correctness, computational binding, and hiding are defined in the usual way.

2.1.1 Ajtai's commitment scheme. We use the standard bounded linear commitment over R_q , matching the compact Ajtai-style commitment of [ACK21, Definition 13]. Fix an input length ℓ_{in} , randomness dimension ℓ_r , output length ℓ_{Com} , and coefficient bound β_{msg} . Setup samples

$$\mathbf{A}_{\text{Com}} \xleftarrow{\$} R_q^{\ell_{\text{Com}} \times \ell_{\text{in}}}$$

and outputs $\text{pp}_{\text{Com}} = (\mathbf{A}_{\text{Com}}, \ell_r, \ell_{\text{Com}}, \beta_{\text{msg}})$. In the blind-signature and ARC protocols, the committed vector has the form

$$\mathbf{w} = [\mu||r_0^H||r_1^H||\bar{r}]^\top,$$

where r_0^H, r_1^H are holder-side rational-hash randomnesses and $\bar{r}$ is commitment-only randomness. The holder computes

$$\mathbf{c} = \mathbf{A}_{\text{Com}} \mathbf{w}.$$

Equivalently, if

$$\mathbf{A}_{\text{Com}} = [A_\mu \mid A_{r_0} \mid A_{r_1} \mid A_{\bar{r}}],$$

then

$$\mathbf{c} = A_\mu \mu + A_{r_0} r_0^H + A_{r_1} r_1^H + A_{\bar{r}} \bar{r}.$$

The verification algorithm accepts an opening iff the committed vector is coefficient-wise bounded by β_{msg} and the displayed linear identity holds.

Definition 2.2 (Module-SIS and Module-ISIS over R_q). Fix integers $n, m \geq 1$, modulus $q \geq 2$, and bound $\beta > 0$.

- M-SIS $_\infty(n, m, q, \beta)$: given $A \in R_q^{n \times m}$, find non-zero $s \in R_q^m$ such that $As = 0$ in R_q^n and $\|s\|_\infty \leq \beta$.
- M-ISIS $_\infty(n, m, q, \beta)$: given $(A, t) \in R_q^{n \times m} \times R_q^n$, find $s \in R_q^m$ such that $As = t$ in R_q^n and $\|s\|_\infty \leq \beta$.

Lemma 2.3 (Binding of the Ajtai commitment). *Let $\mathcal{A}$ be any PPT adversary that opens an Ajtai commitment to two distinct bounded vectors with probability p . Then there is a PPT adversary that solves the corresponding M-SIS $_\infty$ instance with essentially the same probability and runtime.*

PROOF. Two valid openings of the same commitment give two distinct bounded vectors $\mathbf{w}, \mathbf{w}'$ such that $\mathbf{A}_{\text{Com}}(\mathbf{w} - \mathbf{w}') = 0$ in R_q^{Com} . The difference is non-zero and coefficient-wise bounded by $2\beta_{\text{msg}}$, hence it is an SIS solution for the commitment matrix. $\square$

Lemma 2.4 (Statistical hiding of the Ajtai commitment). *Let B_r be the coefficient bound of the commitment-randomness alphabet. If*

$$N\ell_r \log_2(2B_r + 1) \geq N\ell_{\text{Com}} \log_2 q + 2\lambda,$$

then the commitment distribution is statistically $2^{-\lambda}$ -close to a distribution independent of the committed message. In the instantiation above, $B_r = \beta_{\text{msg}}$ is sufficient.

PROOF. This is the direct leftover-hash-lemma argument for the bounded linear Ajtai commitment, specialized to R_q . The commitment randomness contributes $N\ell_r \log_2(2B_r + 1)$ bits of min-entropy and the commitment output space has size $q^{N\ell_{\text{Com}}}$. $\square$

2.2 Lattice trapdoors and rational hashing

We recall only the definitions needed to state the construction. Additional lattice-assumption syntax is deferred to Appendix A.1.

Definition 2.5 (Lattice trapdoor suite). A lattice trapdoor suite over R_q consists of PPT algorithms (Setup, TrapGen, SampPre) such that:

- Setup generates admissible public parameters $(n, m, \beta_{\text{sig}})$.
- $(\mathbf{A}, \text{td}) \leftarrow \text{TrapGen}$ outputs a public matrix $\mathbf{A} \in R_q^{n \times m}$ and a trapdoor, with $\mathbf{A}$ statistically close to uniform over $R_q^{n \times m}$.
- On input (td, T) , SampPre outputs $\mathbf{u} \in R_q^m$ such that $\mathbf{A}\mathbf{u} = T$ in R_q^n and $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$, except with negligible failure probability.

Definition 2.6 (Trapdoor rational hash). A trapdoor rational hash scheme over R_q is a tuple

$$\text{TDRH} = (\text{Setup}, \text{TrapGen}, \text{Eval}, \text{SampPre})$$

consisting of the lattice trapdoor suite above together with a public evaluation algorithm. Its setup also outputs a public hash description B , a randomness space $\mathcal{X}$, and an admissibility error bound δ . For fixed B , the algorithm $\text{Eval}(B, \mu, \chi)$ takes a message μ and witness randomness $\chi \in \mathcal{X}$ and outputs either a target $T \in R_q^n$ or $\perp$.

2.2.1 The BB-tran rational hash. We instantiate the rational hash with the BB-tran form of [DKLW25a, Table 1]. The generic DKLW expression is

$$h_{\mu, \chi}(\tilde{b}^\top) = (u^\top, x_0^\top) \tilde{b}_0 + \frac{1}{\tilde{b}_1 - x_1}.$$

In the notation used in this paper, the public description is $B = (B_0, B_1, B_2, B_3)$, where B_1 and B_2 are public linear maps into R_q . For message μ and randomness (r_0, r_1) , define

$$L_B(\mu, r_0) := B_0 + B_1\mu + B_2r_0.$$

The input is admissible if $B_3 - r_1 \in R_q^\times$. When admissible, let

$$Z = (B_3 - r_1)^{-1}.$$

The BB-tran target is

$$T = h_{\text{tran}, \mu, (r_0, r_1)}(B) := B_0 + B_1\mu + B_2r_0 + Z. \quad (1)$$

Equivalently, the witness form is

$$(B_3 - r_1) \odot Z = 1, \quad T = B_0 + B_1\mu + B_2r_0 + Z, \quad (2)$$

and the eliminated form is

$$(B_3 - r_1) \odot (T - B_0 - B_1\mu - B_2r_0) = 1. \quad (3)$$

The witness form is used in the issuance and showing relations because it separates the inverse check from the linear target equation.

Lemma 2.7 (Eliminated form under admissibility). *Let $R_q \cong \prod_i F_i$ be the CRT decomposition and write $B_3 - r_1 = (d_i)_i$, $T - L_B(\mu, r_0) = (z_i)_i$. If $d_i \neq 0$ for all i , then*

$$T = h_{\text{tran}, \mu, (r_0, r_1)}(B) \iff (B_3 - r_1) \odot (T - B_0 - B_1\mu - B_2r_0) = 1.$$

PROOF. In each CRT slot the defining statement is $z_i = d_i^{-1}$, which is equivalent to $d_i z_i = 1$ when $d_i \neq 0$. Reassembling the CRT slots gives the claim. $\square$

Definition 2.8 (Direct target hiding of BB-tran). We say that the instantiated BB-tran family is directly target-hiding if the following distributions are computationally indistinguishable for all bounded challenge messages μ_0, μ_1 . Sample $b \xleftarrow{\$} \{0, 1\}$, sample r_0, r_1 from the protocol randomness distribution conditioned on $B_3 - r_1 \in R_q^\times$, and output

$$T = B_0 + B_1\mu_b + B_2r_0 + (B_3 - r_1)^{-1}.$$

No PPT distinguisher, given the public parameters and T , can guess b with non-negligible advantage.

The security proof in Section 3 uses this direct target-hiding property as the privacy input for the rational-hash layer. It is separate from the non-blind unforgeability import from DKLW and does not require the target to be computed from the commitment value.

Definition 2.9 (vSIS-style hash-and-sign). Let TDRH be a trapdoor rational hash scheme with parameters $B, \mathcal{X}, \delta, \beta_{\text{sig}}$. The associated signature scheme vSIS = (KG, Sign, Verify) is:

- KG runs $(A, \text{td}) \leftarrow \text{TrapGen}$ and outputs $\text{vk} = A, \text{sk} = \text{td}$.
- Sign(sk, μ) samples $\chi \leftarrow \mathcal{X}$ until $T = \text{Eval}(B, \mu, \chi) \neq \perp$, samples $\mathbf{u} \leftarrow \text{SampPre}(\text{td}, T)$, and outputs $\sigma = (\mathbf{u}, \chi)$.
- Verify(vk, μ, σ) for $\sigma = (\mathbf{u}, \chi)$ checks $\chi \in \mathcal{X}$, $\text{Eval}(B, \mu, \chi) \neq \perp$, $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$, and $A\mathbf{u} = \text{Eval}(B, \mu, \chi)$.

Lemma 2.10 (Imported non-blind security). *Assume the trapdoor parameters are admissible, the BB-tran inadmissibility probability is negligible, the accepted preimages satisfy the norm bound required by [DKLW25a], and the denominator-bound, strong-linear-independence, and predicate side conditions required by the BB-tran analysis of [DKLW25a] hold. Then the non-blind hash-and-preimage signature layer of Definition 2.9 inherits the corresponding DKLW unforgeability guarantees under the stated strong hinted-vSIS/GenISIS assumptions.*

PROOF. This is the specialization of [DKLW25a, Theorems 4, 5, 7, 8 and Corollary 1] to the BB-tran family, together with the BB-tran-specific GenISIS/IntGenISIS lift of [DKLW25a, Section 5.2]. $\square$

2.3 Non-interactive zero-knowledge arguments of knowledge

We use the NIZK-ARK syntax of the SmallWood framework instantiated in the random-oracle model [FR25, Definition 3].

Definition 2.11 (NIZKs in the random oracle model). Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a family of relations. A transparent non-interactive zero-knowledge argument of knowledge for $\mathcal{R}$ in the random-oracle model is a pair $(\text{Prove}^{\text{RO}}, \text{Verify}^{\text{RO}})$ satisfying correctness, straightline-extractable knowledge soundness, and zero knowledge. In particular, zero knowledge means that there exists a simulator Sim^{RO} which, given only the public statement, outputs a proof indistinguishable from a real proof.

We instantiate these proofs using the SmallWood PACS/PIOP framework compiled with Fiat-Shamir in the random-oracle model [FR25].

2.4 Blind signatures

A blind signature scheme allows a holder $\mathcal{H}$ to obtain a signature valid under an issuer's verification key without revealing the signed message to the issuer. We follow the standard syntax of [Fis06].

Definition 2.12 (Blind signature syntax). A blind signature scheme is a tuple

$$\text{BSig} = (\text{Setup}, \text{IssuerKeyGen}, \text{Issue}, \text{Verify})$$

where Setup generates public parameters, IssuerKeyGen generates issuer keys (vk, sk) , Issue is an interactive protocol between issuer and holder, and Verify($\text{vk}, \mu, \text{sig}$) checks signatures.

Definition 2.13 (Correctness). A blind signature scheme is correct if an honestly issued signature verifies except with negligible probability.

Definition 2.14 (Blindness). A blind signature scheme is blind if, for any two messages μ_0, μ_1 , no PPT issuer-side adversary can distinguish the order of two completed issuance transcripts/signatures except with negligible advantage.

Definition 2.15 (One-more unforgeability). A blind signature scheme is one-more unforgeable if any PPT adversary that interacts with the honest issuer in at most Q issuance sessions cannot output $Q + 1$ valid signatures on distinct messages except with negligible probability.

Rate limiting at the ARC layer. We do not introduce a separate stand-alone rate-limiting primitive. In this paper, rate limiting is part of the ARC syntax: the holder presents a credential together with a public nonce and a PRF-derived public tag, and the verifier enforces the policy through state on previously accepted (nonce, tag) pairs.

2.5 Anonymous Rate-Limiting Credentials

Anonymous rate-limited credentials [YWFH26] extend blind signatures by adding a showing algorithm that may create multiple presentations of a previously issued credential.

Definition 2.16 (ARC scheme syntax). Let $\mathcal{N}$ be a finite nonce set. An ARC scheme is a tuple

$$\text{ARC} = (\text{Setup}, \text{IssuerKeyGen}, \text{Issue}, \text{Show}, \text{Verify})$$

where Issue outputs a credential, Show($\text{cred}, \text{nonce}$) outputs a presentation, and Verify($\text{vk}, \text{pres}, \text{state}$) checks the presentation and updates verifier state.

Definition 2.17 (ARC soundness). Let ARC have nonce space $\mathcal{N}$. For $Q = Q(\lambda)$, set $K = Q|\mathcal{N}| + 1$. After at most Q issuance interactions, no PPT adversary can produce K presentations all accepted by a stateful verifier, except with negligible probability.

Definition 2.18 (Presentation unlinkability). An ARC scheme has presentation unlinkability if, for distinct fresh nonces, presentations produced from the same credential are indistinguishable from presentations produced from independently issued credentials, except with negligible advantage.

2.6 Pseudorandom functions

Definition 2.19 (Pseudorandom function). Let $\mathcal{K}, \mathcal{D}, \mathcal{T}$ be sets with $|\mathcal{K}|$ exponential in λ . A function family $F : \mathcal{K} \times \mathcal{D} \rightarrow \mathcal{T}$ is a PRF if oracle access to $F(k, \cdot)$ for uniform $k \xleftarrow{\$} \mathcal{K}$ is computationally indistinguishable from oracle access to a uniform random function.

We instantiate the PRF with a Poseidon2-style permutation [GKS23].

Definition 2.20 (Poseidon2-style permutation). Let q be prime, let $d \geq 3$ satisfy $\gcd(d, q - 1) = 1$, and let $t, R_F, R_P \in \mathbb{N}$ with R_F even. A Poseidon2-style permutation alternates external rounds, which apply the S-box to all lanes, and internal rounds, which apply it to one lane, together with public round constants and invertible linear mixing maps.

Definition 2.21 (Poseidon2 PRF). For key $\mathbf{k} \in \mathbb{F}_q^{\ell_{\text{key}}}$ and public input $\mathbf{n} \in \mathbb{F}_q^{\ell_{\text{in}}}$, define

$$F(\mathbf{k}, \mathbf{n}) = \text{Tr}_{\ell_{\text{tag}}} (P(\mathbf{k} \parallel \mathbf{n}) + (\mathbf{k} \parallel \mathbf{n})),$$

where P is the Poseidon2-style permutation and $\text{Tr}_{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes.

3 BLIND SIGNATURE CONSTRUCTION

This section presents the blind-issuance protocol underlying ARC-SPRUCE. It combines the Ajtai commitment of Section 2.1, the BB-tran rational hash of Section 2.2, and the NIZK model of Section 2.3. The full signed message is $\mu = \mathbf{m} \parallel \mathbf{k}$.

3.1 Relations

The issuance relation binds three objects to one hidden witness: the Ajtai commitment, the centered shared randomness, and the BB-tran target. The public statement is

$$(\text{pp}_{\text{Com}}, \mathbf{c}, T, r_0^I, r_1^I, \mathbf{A}, B),$$

and the hidden witness is

$$(\mu, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z).$$

Define $\mathcal{R}_{\text{Issue}, \text{BS}}$ as the set of all such statement-witness pairs satisfying

$$\begin{aligned} \mathbf{c} &= \mathbf{A}_{\text{Com}} [\mu \parallel r_0^H \parallel r_1^H \parallel \bar{r}]^\top, & \|\mu, r_0^H, r_1^H, \bar{r}, r_0, r_1\|_\infty &\leq \beta_{\text{msg}}, \\ r_0 &= \text{center}(r_0^H + r_0^I), & r_1 &= \text{center}(r_1^H + r_1^I), \\ (B_3 - r_1) \odot Z &= 1, & T &= B_0 + B_1\mu + B_2r_0 + Z. \end{aligned}$$

The signing relation used for a standalone blind signature has public statement $(\mu, \mathbf{A}, B, \beta_{\text{msg}}, \beta_{\text{sig}})$ and hidden witness $(\mathbf{u}, r_0, r_1, Z)$ satisfying

$$\begin{aligned} \|r_0, r_1\|_\infty &\leq \beta_{\text{msg}}, & \|\mathbf{u}\|_\infty &\leq \beta_{\text{sig}}, \\ (B_3 - r_1) \odot Z &= 1, & \mathbf{A}\mathbf{u} &= B_0 + B_1\mu + B_2r_0 + Z. \end{aligned}$$

A standalone blind signature can therefore be represented as a NIZK proof for this second relation. In the ARC layer, the holder instead stores the witness and proves the corresponding relation during showing.

3.2 Protocol

Let $\text{NIZK} = (\text{Prove}, \text{Verify})$ be a NIZK for the above relations.

Setup. Generate the public commitment key $\text{pp}_{\text{Com}} = (\mathbf{A}_{\text{Com}}, \ell_r, \ell_{\text{Com}}, \beta_{\text{msg}})$ and the BB-tran public description $B = (B_0, B_1, B_2, B_3)$. The maps B_1 and B_2 are public linear maps with domains matching the full message μ and the first hash-randomness component r_0 . The public parameters also include the admissibility and bound parameters for the rational hash.

Issuer key generation. The issuer runs $(\mathbf{A}, \text{td}) \leftarrow \text{TrapGen}$ and outputs $\text{vk} = \mathbf{A}$ and $\text{sk} = \text{td}$.

Issuance. The issuer holds sk and the holder has input μ and vk .

- (1) The holder samples $r_0^H, r_1^H, \bar{r}$ coefficient-wise in $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ and sends

$$\mathbf{c} = \mathbf{A}_{\text{Com}} [\mu \parallel r_0^H \parallel r_1^H \parallel \bar{r}]^\top$$

to the issuer.

- (2) The issuer samples r_0^I, r_1^I coefficient-wise in $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ and sends them to the holder.
- (3) The holder computes

$$r_0 = \text{center}(r_0^H + r_0^I), \quad r_1 = \text{center}(r_1^H + r_1^I).$$

If $B_3 - r_1 \notin R_q^\times$, the holder restarts the target derivation. Otherwise it sets

$$Z = (B_3 - r_1)^{-1}, \quad T = B_0 + B_1\mu + B_2r_0 + Z.$$

- (4) The holder sends (T, π_{Issue}) , where π_{Issue} proves membership in $\mathcal{R}_{\text{Issue}, \text{BS}}$.
- (5) The issuer verifies π_{Issue} . If it rejects, the issuer aborts. Otherwise it samples

$$\mathbf{u} \leftarrow \text{SampPre}(\text{td}, T)$$

and sends $\mathbf{u}$ to the holder.

- (6) The holder checks $\mathbf{A}\mathbf{u} = T$ and $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$. If the check fails, it aborts. For a standalone blind signature, the holder outputs a proof π_{Sign} for $\mathcal{R}_{\text{Sign}}$. For ARC issuance, it stores the credential witness $(\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1)$.

Standalone verification. For a standalone signature $\text{sig} = \pi_{\text{Sign}}$ on message μ , the verifier checks the proof for $\mathcal{R}_{\text{Sign}}$. In the ARC construction below, verification is performed through the showing proof instead.

3.3 Correctness and issuer-view hiding

Proposition 3.1 (Correctness). *Assume correctness of the NIZK and of the trapdoor sampler. Then honest issuance produces an accepting standalone signature, and in the ARC variant produces a credential witness satisfying the showing relation.*

PROOF. By construction, the holder computes r_0, r_1, Z, T satisfying the issuance relation. Hence the pre-sign NIZK verifies by completeness. The trapdoor sampler returns $\mathbf{u}$ with $\mathbf{A}\mathbf{u} = T$ and $\|\mathbf{u}\|_\infty \leq \beta_{\text{sig}}$ except with negligible probability, so the post-sign or showing relation is satisfiable. $\square$

Proposition 3.2 (Issuer-view hiding of issuance). *Assume the pre-sign NIZK is zero knowledge, the Ajtai commitment is hiding, the centering map preserves uniformity as in Lemma A.1, and the instantiated BB-tran family satisfies the direct target-hiding property of Definition 2.8. Then the public issuance surface*

$$(\mathbf{c}, r_0^I, r_1^I, T, \pi_{\text{Issue}})$$

is computationally hiding with respect to the holder message μ .

PROOF. First replace the real proof π_{Issue} by a simulated proof for the same public statement. By zero knowledge, this changes the issuer view only negligibly. It remains to analyze $(\mathbf{c}, r_0^I, r_1^I, T)$.

The commitment $\mathbf{c}$ hides the full committed vector, in particular μ, r_0^H, r_1^H , up to the commitment-hiding error. The issuer contributions r_0^I, r_1^I are sampled independently of μ . For each fixed holder contribution r_b^H , Lemma A.1 implies that $\text{center}(r_b^H + r_b^I)$ is uniform over the bounded interval when r_b^I is uniform. Thus the final hash randomness (r_0, r_1) has the distribution required by the direct BB-tran hiding statement, even though r_0^H, r_1^H were committed before the issuer challenge.

Finally, under the direct target-hiding property of BB-tran, the target

$$T = B_0 + B_1\mu + B_2r_0 + (B_3 - r_1)^{-1}$$

hides μ for this randomness distribution, conditioned on admissibility. The admissibility restart depends only on the sampled randomness and the public value B_3 , not on the hidden message. Combining the simulated proof, the hiding commitment, independent issuer randomness, centering uniformity, and target hiding proves the claim. $\square$

Remark 3.3 (Security status). Lemma 2.10 justifies the non-blind hash-and-preimage signature layer. Proposition 3.2 establishes issuer-view hiding of the issuance surface. A full two-session blindness theorem and a one-more unforgeability theorem for the blind-issuance protocol require additional reductions and are not claimed here.

4 FROM BLIND SIGNATURES TO ARC

This section turns the blind-issuance protocol of Section 3 into an anonymous rate-limited credential. The holder signs the full message $\mu = \mathbf{m} \parallel \mathbf{k}$, where $\mathbf{m}$ is the attribute vector and $\mathbf{k}$ is a uniformly sampled secret PRF key.

Credential witness. After successful issuance the holder stores

$$\text{cred} = (\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1),$$

where $\mathbf{Au} = T$ and

$$T = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2r_0 + Z, \quad (B_3 - r_1) \odot Z = 1.$$

The value Z may either be stored or recomputed from r_1 when showing the credential.

4.1 Rate limiting via the signed key

Rate limiting is implemented by deterministic tags derived from the signed secret key $\mathbf{k}$. For a public nonce $\text{nonce} \in \mathcal{D}$, the holder computes

$$\text{tag} = F(\mathbf{k}, \text{nonce}).$$

If the holder reuses a nonce, the tag repeats and the verifier can reject by consulting local state. For fresh nonces, PRF security makes the tags computationally independent of previous tags. Since $\mathbf{k}$ is part of the signed message, the holder cannot change the PRF key without forging a new credential.

4.2 ARC protocol

The setup and issuer-key-generation algorithms are those of Section 3.2.

Issue. The holder executes the issuance protocol on message $\mu = \mathbf{m} \parallel \mathbf{k}$. If the final check $\mathbf{Au} = T$ succeeds, it stores $\text{cred} = (\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1)$.

Show. On input cred and nonce nonce , the holder computes

$$Z = (B_3 - r_1)^{-1}, \quad \text{tag} = F(\mathbf{k}, \text{nonce}),$$

and produces a NIZK proof π_{show} of knowledge of

$$\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z$$

such that

$$(B_3 - r_1) \odot Z = 1,$$

$$\mathbf{Au} = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2r_0 + Z,$$

$$\text{tag} = F(\mathbf{k}, \text{nonce}),$$

and all hidden values satisfy the relevant bounds. It outputs

$$\text{pres} = (\text{nonce}, \text{tag}, \pi_{\text{show}}).$$

Verify. The verifier checks π_{show} against the public statement containing $(\text{vk}, B, \text{nonce}, \text{tag})$ and the public bounds. If the proof rejects, verification returns reject. If $(\text{nonce}, \text{tag})$ is already present in verifier state, verification also rejects. Otherwise the verifier stores $(\text{nonce}, \text{tag})$ and accepts.

4.3 Concrete PRF instantiation

We instantiate the PRF using the Poseidon2-like construction of Definition 2.21. Algorithm 1 summarizes the computation.

Algorithm 1 $F(\mathbf{k}, \mathbf{n})$: Poseidon2-like with feed-forward and truncation

```

1:  $x \leftarrow \mathbf{k} \parallel \mathbf{n}; x^{(0)} \leftarrow x$ 
2: for  $i = 1$  to  $R_F/2$  do
3:    $x \leftarrow E_i(x)$ 
4: for  $i = 1$  to  $R_P$  do
5:    $x \leftarrow I_i(x)$ 
6: for  $i = R_F/2 + 1$  to  $R_F$  do
7:    $x \leftarrow E_i(x)$ 
8:  $y \leftarrow x + x^{(0)}$ 
9: return  $\text{Tr}_{\text{tag}}(y)$ 
```

In the showing proof, the equation $\text{tag} = F(\mathbf{k}, \text{nonce})$ is enforced with the checkpointed arithmetization described in Section 5.4 and Appendix E.

4.4 Security decomposition

The ARC layer combines the following ingredients:

- (1) the non-blind signature security imported in Lemma 2.10;
- (2) issuer-view hiding of the issuance surface from Proposition 3.2;
- (3) one-more unforgeability of the blind-issuance protocol;
- (4) knowledge soundness and zero knowledge of the showing NIZK;
- (5) correctness and pseudorandomness of the PRF; and
- (6) verifier-side state recording previously accepted $(\text{nonce}, \text{tag})$ pairs.

The first two ingredients are established in the main text under the stated assumptions. The third remains a separate reduction goal. The last item is an application-layer enforcement rule rather than a standalone cryptographic assumption.

Conditional policy soundness. Knowledge soundness of the showing proof yields a witness satisfying the BB-tran signature relation and the PRF relation. Assuming one-more unforgeability of the blind-issuance layer, an adversary cannot obtain more valid credential witnesses than issuance sessions allow. Since $\text{tag} = F(\mathbf{k}, \text{nonce})$ is uniquely determined by the signed key and nonce, a verifier

that records spent (nonce, tag) pairs rejects same-nonce replays deterministically.

Conditional presentation unlinkability. For distinct fresh nonces, zero knowledge hides the credential witness and PRF security makes the public tags pseudorandom. Presentation unlinkability also relies on issuance-layer privacy because the adversary may control the issuer during issuance. Under Proposition 3.2, showing transcripts are unlinkable up to the intended nonce-reuse caveat.

Attribute privacy and selective disclosure. The message part $\mathbf{m}$ can encode attributes. The showing NIZK can be extended to prove predicates over $\mathbf{m}$ or to selectively reveal components without changing the issuance protocol.

5 SMALLWOOD PROGRAMMING MODEL

This section records the semantic SmallWood statements used by ARC-SPRUCE. The concrete row layouts, DECS/LVCS/PCS openings, and proof-stack details are deferred to Appendix C. The prover commits to packed witness rows over a set Ω and proves PACS-style constraints on those committed rows.

5.1 Proof interface

Fix a packing set $\Omega = \{\omega_1, \dots, \omega_s\} \subset \mathbb{F}_q$ and a disjoint hiding set $\Omega' \subset \mathbb{F}_q \setminus \Omega$. A compiled witness is a matrix of row values over $\mathbb{F}_q$. Each row is interpolated to a low-degree polynomial whose values on Ω encode the witness and whose values on Ω' are random masks. SmallWood then checks parallel constraints pointwise over Ω and, when needed, aggregated constraints through Ω -sum identities.

5.2 Issuance as a SmallWood statement

The public issuance statement is

$$(\text{ppCom}, \mathbf{c}, T, r_0^I, r_1^I, \mathbf{A}, B),$$

and the hidden witness is

$$(\mathbf{m}, \mathbf{k}, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z).$$

Semantically, the proof establishes four claims. First, the commitment opens to the same message and holder-side randomness used later:

$$\mathbf{c} = \mathbf{A}_{\text{Com}} [\mathbf{m} \parallel \mathbf{k} \parallel r_0^H \parallel r_1^H \parallel \bar{r}]^\top.$$

Second, the public issuer randomness is combined correctly with the committed holder randomness:

$$r_0 = \text{center}(r_0^H + r_0^I), \quad r_1 = \text{center}(r_1^H + r_1^I).$$

Third, the inverse witness is valid:

$$(B_3 - r_1) \odot Z = 1.$$

Fourth, the public target is the BB-tran target for the same hidden message and centered randomness:

$$T = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2 r_0 + Z.$$

All message and randomness rows are constrained to lie in the coefficient interval $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$.

The corresponding committed rows consist of carrier rows for the packed message, the holder-side randomness, commitment-only randomness, centered randomness, carry rows for the centering map, and the inverse witness Z . The pre-sign proof is dominated by

commitment linearity, centering, range membership, the inverse-witness equation, and the target identity above.

5.3 Showing as a SmallWood statement

The public showing statement is

$$(\text{nonce}, \text{tag}, \mathbf{A}, B, \beta_{\text{msg}}, \beta_{\text{sig}})$$

with verifier state handled outside the proof. The hidden witness is

$$(\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z).$$

The proof certifies

$$\begin{aligned} (B_3 - r_1) \odot Z &= 1, \\ \mathbf{A}\mathbf{u} &= B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2 r_0 + Z, \\ \text{tag} &= F(\mathbf{k}, \text{nonce}), \end{aligned}$$

with coefficient bounds on $\mathbf{m}, \mathbf{k}, r_0, r_1$ and the public shortness bound on $\mathbf{u}$.

In the compiled statement, the signature surface proves shortness of $\mathbf{u}$ and binds the derived target to $\mathbf{A}\mathbf{u}$. The BB-tran part carries source rows for $T = \mathbf{A}\mathbf{u}, \mathbf{m} \parallel \mathbf{k}, r_0, r_1$, and Z . The PRF companion rows bind the same hidden key $\mathbf{k}$ to the public tag.

5.4 PRF companion relation

The PRF relation ties the public tag to the same hidden key that appears in the signed message. Selector constraints bind the PRF key lanes to $\mathbf{k}$; checkpoint constraints certify the nonlinear S-box steps of the Poseidon2-like permutation; and the final feed-forward and truncation constraints bind the resulting lanes to (nonce, tag).

5.5 Statement strength

The showing statement certified by SmallWood is the coefficient-domain BB-tran identity in witness form:

$$(B_3 - r_1) \odot Z = 1, \quad \mathbf{A}\mathbf{u} = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2 r_0 + Z.$$

Equivalently, it implies the eliminated identity

$$(B_3 - r_1) \odot (\mathbf{A}\mathbf{u} - B_0 - B_1(\mathbf{m} \parallel \mathbf{k}) - B_2 r_0) = 1.$$

Thus any accepting showing proof yields, by knowledge soundness, a witness satisfying the abstract BB-tran signature relation and the PRF-tag relation.

Lemma 5.1 (Witness consequence of an accepting showing proof). *If the post-sign NIZK is knowledge sound for the compiled showing relation, then every accepting proof yields a witness $(\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z)$ satisfying*

$$\mathbf{A}\mathbf{u} = h_{\text{tran}, \mathbf{m} \parallel \mathbf{k}, (r_0, r_1)}(B)$$

and $\text{tag} = F(\mathbf{k}, \text{nonce})$.

PROOF. The extractor returns a witness satisfying the compiled equations above. The inverse equation gives $Z = (B_3 - r_1)^{-1}$ under admissibility, and substituting it into the target equation gives the abstract BB-tran relation. $\square$

6 PARAMETERS AND INTEGRATION

This section explains how parameters interact across the ARC-SPRUCE stack: the Ajtai commitment, the BB-tran rational hash, the lattice preimage sampler, SmallWood, and the PRF. Detailed parameter tables and extended benchmark data are deferred to Appendix D.

6.1 Parameter dependencies

All components share the same base ring R_q and modulus q :

- the commitment matrix, signature map $\mathbf{A}$, and hash key B are defined over R_q ;
- SmallWood constraints are evaluated over the base field $\mathbb{F}_q$ and therefore use the same modulus;
- the PRF is instantiated over $\mathbb{F}_q$ so that its constraints are native to the proof system.

The coefficient bound β_{msg} governs the message space, holder-side randomness, issuer-side randomness, centering map, and membership polynomials in the NIZKs.

6.2 Commitment and shared-randomness parameters

The Ajtai commitment binds the vector

$$[\mu \| r_0^H \| r_1^H \| \bar{r}]^\top.$$

Its hiding parameters must satisfy the leftover-hash condition of Lemma 2.4. The commitment-only randomness $\bar{r}$ and any additional committed randomness coordinates must provide enough min-entropy to statistically hide the full committed vector.

The centering map uses the interval $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ and modulus

$$\Delta = 2\beta_{\text{msg}} + 1.$$

For correctness of the shared-randomness step, both holder and issuer contributions must lie in $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ coefficient-wise. Lemma A.1 is then applicable coefficient-wise: for fixed holder randomness, the centered sum is uniform when the issuer contribution is uniform.

6.3 Sampler parameters

The trapdoor sampler must output short signatures $\mathbf{u}$ with overwhelming probability and with a distribution close to a discrete Gaussian over the appropriate lattice coset. This requires choosing:

- ring degree N and modulus q ,
- trapdoor quality,
- Gaussian widths for the sampler, and
- an acceptance predicate that yields the public verification bound β_{sig} .

We use an NTRU-style trapdoor together with a Gaussian preimage sampler; details and tuning rules are given in Appendix B.

6.4 BB-tran rational-hash parameters

The BB-tran target is evaluated only when

$$B_3 - r_1 \in R_q^\times.$$

Honest target derivation therefore restarts when this condition fails. The restart probability must be negligible or accounted for as the rational-hash admissibility error δ . Since r_1 is the centered sum

of holder and issuer contributions, its distribution is the centered bounded distribution required by the target-hiding statement.

The pre-sign proof certifies the witness form

$$(B_3 - r_1) \odot Z = 1, \quad T = B_0 + B_1\mu + B_2r_0 + Z.$$

The showing proof certifies the same relation with $T = \mathbf{A}\mathbf{u}$. Equivalently, it certifies

$$(B_3 - r_1) \odot (\mathbf{A}\mathbf{u} - B_0 - B_1\mu - B_2r_0) = 1.$$

Membership checks for $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ are implemented via vanishing polynomials over $\mathbb{F}_q$, so one requires $\beta_{\text{msg}} \ll q$ to avoid modular wrap-around artifacts.

Non-blind reduction parameters. The non-blind reduction imported from DKLW25 is parameterized by a norm bound β_{DKLW} , an inadmissibility bound δ , and the predicate/linear-independence side conditions of the BB-tran family. The public verifier bound β_{sig} should therefore dominate the relevant DKLW norm bound under the chosen norm comparison.

6.5 SmallWood parameters

SmallWood introduces parameters controlling packing, masking, and soundness:

- the packing size $s = |\Omega|$,
- the tail length ℓ for HVZK masking and degree enforcement,
- the spot-check count ℓ' and the number of masks ρ for PIOP soundness,
- the degree bounds for witness rows and constraint-derived masks, and
- the small-field knob θ when using an extension-field variant.

The pre-sign proof is dominated by the Ajtai opening, centering equations, inverse-witness check, and BB-tran target identity. The showing proof adds the signature shortness gadget, the BB-tran signature equation with $T = \mathbf{A}\mathbf{u}$, and the PRF checkpoint constraints.

6.6 PRF parameters

The PRF must be secure at the target security level, efficiently representable in constraints, and compatible with $\mathbb{F}_q$. We use a Poseidon2-like permutation with feed-forward and truncation. Its parameters include the state width t , round counts (R_F, R_P) , MDS matrices, round constants, S-box exponent d , and truncation length ℓ_{tag} . We require $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$ so the tag contains λ bits of entropy.

6.7 Sizing discussion

Concrete proof sizes depend jointly on the sampler parameters, the SmallWood masking parameters, the BB-tran relation, and the PRF checkpoint schedule. The issuance proof is smaller because the target T is public and the statement only checks consistency of the committed witness with that target. The showing proof additionally proves knowledge of a short preimage and a correct PRF evaluation.

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A ADDITIONAL PRELIMINARIES

A.1 Security assumptions for Lattices

Fix a public left factor F , hint space H , target space G , predicate P , and query bound Q .

- **Hint-vSIS.** The adversary chooses a query set $Q \subseteq H$ and a fresh target function $g^* \in G \setminus Q$ before seeing the sampled matrix A . It then receives short preimages of the queried hints $q(A)$ under $F(A)$ for every $q \in Q$, and must output a non-zero short vector u^* such that $F(A)u^* = g^*(A)$. The instance is restricted by the predicate condition $P(F \cup Q \cup (\{g^*\} \setminus \{0\})) = 1$.
- **s-Hint-vSIS.** This is the strong variant of the same experiment. The adversary still chooses the hint set Q before seeing A , but it is allowed to choose the fresh target g^* only after seeing A and the hinted preimages.
- **s-\$-Hint-vSIS.** This is the strong random-hinted variant. It is the same as s-Hint-vSIS, except that the query set Q is sampled uniformly at random from H rather than chosen by the adversary.

Now fix a keyed evaluation family $f : K \times M \times X \rightarrow R_q^n$.

- **GenSIS_f.** The challenger samples a public key $\kappa \leftarrow K$ and a uniform matrix $A \xleftarrow{\$} R_q^{n \times m}$. It provides a preimage oracle that, on input $\mu \in M$, samples $\chi \leftarrow X$, sets $t = f(\kappa, \mu, \chi)$, returns a short preimage s such that $As = t$, and records the tuple (s, μ, χ) . The adversary wins if it outputs a fresh tuple (s^*, μ^*, χ^*) not previously returned by the oracle such that $As^* = f(\kappa, \mu^*, \chi^*)$ and $0 < \|s^*\| \leq \beta$.
- **IntGenSIS_f.** This is an interactive version where the challenger offers a signing oracle with an additional linear masking term.

[DKLW25b] shows how this interactive problem reduces back to GenSIS_f under increased parameters.

A.2 Centering map and carries

Let $\text{BoundB} \in \mathbb{Z}_{>0}$ and $\Delta = 2\text{BoundB} + 1$. Define the centering map $\text{center} : [-2\text{BoundB}, 2\text{BoundB}] \rightarrow [-\text{BoundB}, \text{BoundB}]$ by

$$\text{center}(x) = \begin{cases} x & \text{if } x \in [-\text{BoundB}, \text{BoundB}], \\ x + \Delta & \text{if } x \in [-2\text{BoundB}, -\text{BoundB} - 1], \\ x - \Delta & \text{if } x \in [\text{BoundB} + 1, 2\text{BoundB}]. \end{cases}$$

Applied coefficient-wise, this map allows unbiased combination of two bounded values when at least one is uniform.

Lemma A.1 (Centering preserves uniformity). *Fix any $a \in [-\text{BoundB}, \text{BoundB}]$.*

If $b \xleftarrow{\$} [-\text{BoundB}, \text{BoundB}]$ then $\text{center}(a + b)$ is uniform over $[-\text{BoundB}, \text{BoundB}]$.

A.3 Vanishing polynomials for membership constraints

Membership constraints are expressed using vanishing polynomials over $\mathbb{F}_q$ with signed lifting.

Definition A.2 (Vanishing polynomial for $[-\text{BoundB}, \text{BoundB}]$). Let $\text{BoundB} \in \mathbb{Z}_{>0}$. Define

$$P_{\text{BoundB}}(X) = \prod_{i=-\text{BoundB}}^{\text{BoundB}} (X - \langle i \rangle_q) \in \mathbb{F}_q[X].$$

For $x \in \mathbb{F}_q$, we have $x \in [-\text{BoundB}, \text{BoundB}]$ (via signed lifting) iff $P_{\text{BoundB}}(x) = 0$ in $\mathbb{F}_q$.

Definition A.3 (Vanishing polynomial for $\{-1, 0, 1\}$). Define $P_1(X) = (X + 1)X(X - 1) \in \mathbb{F}_q[X]$. Then $P_1(x) = 0$ iff $x \in \{-1, 0, 1\}$.

B GAUSSIAN PREIMAGE SAMPLING

This appendix details the Gaussian preimage sampler used by the issuer to sign a target $t \in R_q^n$ by producing a short preimage $u \in R^m$ such that $Au = t$ in R_q^n . Our instantiation follows the vSIS-BB-tran setting [DKLW25a] and relies on an NTRU trapdoor generated via Antrag together with a hybrid two-plane sampler [ENS⁺23]. In the concrete ARC-SPRUCE instantiation, the signed target is the single-coordinate h_{tran} value produced from the message and centered shared randomness.

B.1 The preimage sampling problem

Given a public linear map $A \in R_q^{n \times m}$ and a target $t \in R_q^n$, the signer samples

$$u \leftarrow \text{SampPre}(\text{td}_A, t) \quad \text{such that} \quad Au \equiv t \pmod{q}$$

and u is short (according to a public bound β_{sig}). Equivalently, u lies in the coset lattice

$$\Lambda_{\perp}^t(A) := \{x \in R^m : Ax \equiv t \pmod{q}\}.$$

We require: (i) correctness ($Au = t$ and the norm test fails only with negligible probability), (ii) (approx.) Gaussianity of u over the coset (statistical hiding of the trapdoor), and (iii) a tight tail bound that supports efficient ZK range/norm checks in the post-sign proof.

B.2 NTRU trapdoor form and quality

We work with the rank-2 NTRU module over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ and its quotient R_q . An NTRU public key has the form $h = gf^{-1} \bmod q$ for secret $f, g \in R$. The corresponding module is

$$L_{\text{NTRU}}(h) = \{(u, v) \in R^2 : uh - v \equiv 0 \pmod{q}\}.$$

A trapdoor basis of $L_{\text{NTRU}}(h)$ is

$$B_{f,g} = \begin{pmatrix} f & F \\ g & G \end{pmatrix}, \quad \text{with} \quad fG - gF = q,$$

whose columns are short and satisfy $g/f \equiv G/F \equiv h \pmod{q}$. Following [ENS⁺23], we measure trapdoor quality via per-slot energies under the canonical embeddings φ_i , requiring a uniform window around q . Concretely, for a quality parameter $\alpha > 1$ we target

$$\frac{q}{\alpha^2} \leq |\varphi_i(f)|^2 + |\varphi_i(g)|^2 \leq \alpha^2 q \quad \text{for all slots } i.$$

Smaller α corresponds to tighter trapdoors and, for fixed N, q , to shorter signatures for the same security.

B.3 Sampler definition and tuning

Our sampler is a ring-based hybrid two-plane sampler in the evaluation domain. The sampler's main tuning parameters are:

- modulus q and ring degree N ,
- trapdoor quality α ,
- acceptance radius R_{acc} (controls rejections and ℓ_∞/ℓ_2 tails),
- per-slot widths $\sigma_k[i]$ (two planes $k \in \{1, 2\}$).

Per-slot calibration. Let $\tilde{b}_k[i]$ denote the per-slot Gram-Schmidt lengths for plane k at slot i . A standard calibration equalizes acceptance probability across slots:

$$\sigma_k[i] = \sqrt{\frac{\alpha^2 q R_{\text{acc}}^2}{\|\tilde{b}_k[i]\|^2} - R_{\text{acc}}^2} \quad (k \in \{1, 2\}).$$

This prevents weak slots and yields consistent output variance.

Hiding threshold (smoothing). To achieve statistical hiding, we enforce a floor $\sigma_{\min}$ such that $\sigma_k[i] \geq \sigma_{\min}$ for all k, i , where $\sigma_{\min}$ exceeds the smoothing parameter of the relevant lattice cosets. In practice, we apply a global scale factor to raise all $\sigma_k[i]$ above the theoretical floor with slack.

B.4 Annulus sampling (Antrag key generation)

Antrag replaces trial-and-error trapdoor generation with direct annulus sampling in the Fourier domain [ENS⁺23]. For each conjugate pair (there are $N/2$ independent slots), the key generator samples $(|\varphi_i(f)|, |\varphi_i(g)|)$ uniformly in an annulus arc $A^+(r, R)$ with

$$\sqrt{q}/\alpha < r < R < \alpha\sqrt{q},$$

chooses random phases, inverse-embeds to the coefficient domain, and rounds to obtain $(f, g) \in R^2$. By selecting a “middle” annulus window (see [ENS⁺23]) one ensures that the rounded trapdoor satisfies the desired α -window with high probability and with modest repetition.

B.5 Hybrid two-plane sampler (details)

We sketch the two-plane hybrid sampler in the notation of [ENS⁺23]. Let td_A encode an NTRU trapdoor for A and let $t \in R_q^n$ be the target. Write $\langle \cdot \rangle_q$ for signed coefficient lift and define center_q as coefficient-wise centering into $(-\frac{q}{2}, \frac{q}{2}]$.

Algorithm 2 SampPre(td_A, t): two-plane hybrid sampler (sketch)

- 1: Lift t to coefficients and set $c \leftarrow \text{center}_q(t) \in (-\frac{q}{2}, \frac{q}{2}]^n$.
 - 2: Using td_A , compute per-slot widths $\sigma_k[i]$ (two planes) calibrated to trapdoor quality.
 - 3: **repeat**
 - 4: Sample slot-wise Gaussian pairs on the two planes, reconstruct a coefficient-domain candidate u .
 - 5: Accept iff the norm predicate holds for u (e.g., ℓ_2 and/or ℓ_∞ bound).
 - 6: **until** accepted
 - 7: **return** u .
-

Public relations. Downstream verification and proofs use only the public relation $Au = t$ and shortness of u (bounded by β_{sig}), checked by the holder on receipt and enforced by the issuer's rejection sampling. The post-sign NIZK then proves knowledge of such a short u without revealing it.

C SMALLWOOD PROOF-STACK DETAILS

This appendix records the compiled surfaces used by the issuance and showing statements. It is descriptive: the semantic statements are those of Section 5.

C.1 Row inventories

Issuance rows. The issuance proof commits rows encoding:

$$\mathbf{m}, \mathbf{k}, r_0^H, r_1^H, \bar{r}, r_0, r_1, Z,$$

together with carry rows for the centering equations and auxiliary rows for range membership. The active relations are

$$\begin{aligned} \mathbf{c} &= \mathbf{A}_{\text{Com}}[\mathbf{m} \|\mathbf{k}\| r_0^H \|r_1^H \|\bar{r}]^\top, \\ r_0 &= \text{center}(r_0^H + r_0^I), \quad r_1 = \text{center}(r_1^H + r_1^I), \\ (B_3 - r_1) \odot Z &= 1, \\ T &= B_0 + B_1(\mathbf{m} \|\mathbf{k}\|) + B_2 r_0 + Z. \end{aligned}$$

Showing rows. The showing proof commits rows encoding:

$$\mathbf{u}, \mathbf{m}, \mathbf{k}, r_0, r_1, Z,$$

plus rows for the signature shortness gadget and the PRF companion relation. The active algebraic relations are

$$\begin{aligned} (B_3 - r_1) \odot Z &= 1, \\ \mathbf{A}\mathbf{u} &= B_0 + B_1(\mathbf{m} \|\mathbf{k}\|) + B_2 r_0 + Z, \\ \text{tag} &= F(\mathbf{k}, \text{nonce}). \end{aligned}$$

C.2 Commitment and centering constraints

The Ajtai opening constraint is linear over R_q and is compiled coefficient-wise into SmallWood rows. The centering map is represented by carry variables. With $\Delta = 2\beta_{\text{msg}} + 1$, each coefficient

satisfies

$$r_b = r_b^H + r_b^I - \Delta j_b, \quad j_b \in \{-1, 0, 1\},$$

for $b \in \{0, 1\}$. Membership in the bounded interval and in $\{-1, 0, 1\}$ is enforced by the vanishing polynomials of Appendix A.

C.3 BB-tran constraints

The BB-tran constraints are expressed in witness form. Issuance checks the public target equation

$$T = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2 r_0 + Z$$

and the inverse equation

$$(B_3 - r_1) \odot Z = 1.$$

Showing uses the same inverse equation and replaces the public target by the signature surface $\mathbf{Au}$:

$$\mathbf{Au} = B_0 + B_1(\mathbf{m} \parallel \mathbf{k}) + B_2 r_0 + Z.$$

Equivalently, the showing proof implies the eliminated coefficient-domain relation

$$(B_3 - r_1) \odot (\mathbf{Au} - B_0 - B_1(\mathbf{m} \parallel \mathbf{k}) - B_2 r_0) = 1.$$

C.4 PRF companion constraints

The PRF companion relation binds the public pair (nonce, tag) to the hidden signed key $\mathbf{k}$. The proof commits key-lane rows, selected S-box checkpoint rows, and any auxiliary rows needed to replay the Poseidon2-style linear layers between checkpoints. Selector constraints tie the key-lane rows to the signed key coordinates; final feed-forward and truncation constraints tie the public tag to the resulting PRF output.

C.5 Single committed witness

All rows in one proof live under one SmallWood commitment. This ensures that the prover cannot use one hidden message for the Ajtai opening, a different message for the BB-tran target, and a third key for the PRF. In the showing proof, the same hidden key $\mathbf{k}$ is used simultaneously in the signature relation and the tag relation.

D EXTENDED PARAMETERS AND BENCHMARKS

This appendix provides extended parameter discussion and additional benchmark data.

D.1 Admissibility checklist

A parameter tuple is *admissible* for ARC-SPRUCE if it simultaneously satisfies:

- (1) **Bound vs. modulus:** $\beta_{\text{msg}} \ll q$ so membership checks on $[-\beta_{\text{msg}}, \beta_{\text{msg}}]$ do not wrap modulo q .
- (2) **Sampler hiding and shortness:** Gaussian widths exceed the required smoothing threshold and the signature bound β_{sig} holds with overwhelming probability.
- (3) **Rational-hash guard:** denominator invertibility succeeds with negligible failure probability under the holder randomness distribution.
- (4) **SmallWood degree/soundness:** the DECS/LVCS degree bounds and PIOP soundness parameters meet the target security λ .

- (5) **PRF entropy:** truncation length ℓ_{tag} yields at least λ bits of tag entropy.

D.2 Smoothing and sampler budgets

Let $\eta_\epsilon(\Lambda)$ denote the smoothing parameter of a lattice Λ . A sufficient condition for statistical hiding in Gaussian preimage sampling is to choose Gaussian widths above a floor $\sigma_{\min} \geq \eta_\epsilon(\Lambda_\perp(A))$, with slack (Appendix B). For the ambient lattice $\mathbb{Z}^{2N}$ one may use the classical bound

$$\eta_\epsilon(\mathbb{Z}^{2N}) = \sqrt{\ln(4N(1 + 2^\lambda))}/\pi.$$

In our tuning, trapdoor quality α and acceptance parameters are chosen so that the output shortness bound β_{sig} holds with overwhelming probability.

D.3 Example lattice parameters

Table 1 records one concrete instantiation compatible with the sizing discussion in Section 6.

Knob / Result	Symbol	Value (example)
Ring degree	N	1024
Modulus	q	1038337
Security target	λ	128
Trapdoor quality	α	≈ 1.23 (measured)
Gaussian width scale	σ	tuned per-slot (Appendix B)
Signature bound	β_{sig}	derived from sampler tail bound

Table 1: Example lattice parameters (illustrative; see Appendix B for the sampling interface and tuning).

D.4 SmallWood soundness terms

SmallWood's soundness is a combination of degree soundness (DECS), LVCS binding, and PIOP sampling errors [FR25]. For parameters $(N, s, \ell, \ell', \rho, \eta)$ and degree bounds $d_{\text{decs}} = s + \ell - 1$ and d_Q , typical upper bounds take the form:

$$\epsilon_{\text{decs}} \lesssim \binom{N}{d_{\text{decs}} + 2} q^{-\eta}, \quad \epsilon_{\text{sum}} = q^{-\rho}, \quad \epsilon_{\text{tail}} \approx \frac{\binom{d_Q}{\ell'}}{\binom{|S|}{\ell'}}.$$

Our issuance statement uses only linear constraints (given public T), so d_Q is driven by membership checks; the showing statement additionally depends on the PRF effective degree (Appendix E).

D.5 SmallWood knobs (example)

For the issuance and showing proofs discussed in Section 6, one example parameter tuple is:

$$(s, \ell, \ell', \rho, \theta, \eta) = (16, 25, 2, 2, 4, 19),$$

with masking degree d_Q derived from the instantiated constraint degrees. The pre-sign statement remains linear in the witness (given public T), while the post-sign statement's effective degree is dominated by the PRF S-box composition.

D.6 Extended benchmark grid (SmallWood)

For additional context on how SmallWood knobs affect proof size and proving time, Table 2 reports an example sweep over packing and soundness parameters (small-field variant).

ProofKB	Time (s)	Bits	NCols	ℓ	ℓ'	ρ	θ	η
9.32	0.266	133.44	6	22	2	2	4	17
10.07	0.228	146.93	4	24	2	2	4	17
11.52	0.250	133.22	4	28	2	2	4	17
14.06	0.630	198.01	4	34	8	5	2	26
14.17	0.654	192.26	6	34	8	5	2	26
14.24	0.670	198.48	4	34	8	6	2	26

Table 2: Example sweep over SmallWood knobs (proof size, prover time, and soundness bits). Sizes exclude public parameters.

E PRF DETAILS AND MISCELLANEOUS

This appendix records a concrete ZK-friendly PRF instantiation compatible with the canonical ARC protocol.

E.1 PRF specification

We use a Poseidon2-like permutation with feed-forward and truncation. Let $k \in \mathbb{F}_q^{\ell_{\text{key}}}$ be the secret key (derived from the signed $\mathbf{k}$) and $n \in \mathbb{F}_q^{\ell_{\text{nonce}}}$ encode the public nonce. Let $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ be the permutation width. Define

$$F(k, n) = \text{Tr}(P(k||n) + (k||n)),$$

where $\text{Tr} : \mathbb{F}_q^t \rightarrow \mathbb{F}_q^{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes and P is a Poseidon2-style permutation.

E.2 Permutation structure

Let $d \geq 3$ be the smallest constant such that $\gcd(d, q-1) = 1$. Fix round counts R_F (external rounds, even) and R_P (internal rounds), round constants, and MDS matrices M_E, M_I . The permutation has the form

$$P = E_{R_F} \circ \dots \circ E_{R_F/2+1} \circ I_{R_P} \circ \dots \circ I_1 \circ E_{R_F/2} \circ \dots \circ E_1.$$

Each external round applies the S-box to all lanes, while each internal round applies the S-box to lane 1 only. This structure reduces nonlinear constraints in the post-sign NIZK while retaining diffusion.

External and internal rounds. Let $x = (x_1, \dots, x_t) \in \mathbb{F}_q^t$. For each external round $i \in [1, R_F]$, fix per-lane constants $(c_1^{(i)}, \dots, c_t^{(i)}) \in \mathbb{F}_q^t$. For each internal round $i \in [1, R_P]$, fix a scalar constant $\hat{c}^{(i)} \in \mathbb{F}_q$. Then:

$$E_i(x) = M_E \cdot ((x_1 + c_1^{(i)})^d, \dots, (x_t + c_t^{(i)})^d)^\top,$$

$$I_i(x) = M_I \cdot ((x_1 + \hat{c}^{(i)})^d, x_2, \dots, x_t)^\top.$$

The matrices $M_E, M_I \in \mathbb{F}_q^{t \times t}$ are chosen so that they are invertible and provide sufficient diffusion (e.g., Cauchy-style MDS matrices as in Poseidon2).

E.3 Parameter generation

All PRF parameters are public. A typical parameter generator fixes:

- the base field modulus q (shared with the rest of the protocol),
- the S-box exponent d with $\gcd(d, q-1) = 1$,
- width $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ and truncation ℓ_{tag} ,
- round counts R_F, R_P ,
- MDS matrices M_E, M_I and round constants $\{c^{(i)}\}, \{\hat{c}^{(i)}\}$.

The truncation length is chosen to satisfy $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$, ensuring that the tag contains λ bits of entropy.

E.4 Arithmetization strategy (System A with checkpoints)

Status of the checkpoint schedule. Section 5.4 uses only the abstract interface of three PRF companion families: key binding, non-linear checkpoint constraints, and final feed-forward/truncation constraints. The grouped schedule recorded in this appendix is the concrete realization used for the present instantiation and measurements. Alternative checkpoint schedules with the same interface could be substituted without changing the statement layer of Section 5.

To keep showing proofs compact, we commit only selected non-linear checkpoints:

- **Key lanes.** We commit one row per key lane $\text{Key}[i]$ and enforce equality to an encoding of signed $\mathbf{k}$ using selector constraints at fixed Ω -coordinates.
- **S-box outputs.** We commit only the S-box outputs at a checkpoint schedule (e.g., every internal round and periodically among external rounds). Linear wiring between checkpoints (add-round-constants, MDS multiplications) is recomputed by the verifier from opened evaluations and public constants.

This yields a System-A style constraint system where the dominant degree comes from composing at most a small number of unchecked S-box layers between checkpoints.